



Started on	Wednesday, 13 August 2025, 10:37 AM
State	Finished
Completed on	Friday, 15 August 2025, 12:03 PM
Time taken	2 days 1 hour
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2  int c=0;
3  void reverse(int n)
4  {
5      int rev = 0, remainder;
6      c++;
7      while (n != 0)
8      {
9          c++;
10         remainder = n % 10;
11         c++;
12         rev = rev * 10 + remainder;
13         c++;
14         n /= 10;
15         c++;
16     }
17 }
18
19
20 int main()
21 {
22     int n;
23     scanf("%d",&n);
24     reverse(n);
25     printf("%d",c+2);
26     return 0;
27 }
```

	Input	Expected	Got	
✓	12	11	11	✓

	Input	Expected	Got	
✓	1234	19	19	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Wednesday, 13 August 2025, 10:37 AM
State	Finished
Completed on	Wednesday, 13 August 2025, 11:29 AM
Time taken	51 mins 38 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2  int c=0;
3  void function(int n)
4  {
5      for(int i=n/2; i<n; i++){
6          c++;
7          c++;
8          for(int j=1; j<n; j = 2 * j){
9              c++;
10             c++;
11             for(int k=1; k<n; k = k * 2){
12                 c++;
13                 c++;
14             }
15         }
16     }
17 }
18
19 int main()
20 {
21     int n;
22     scanf("%d",&n);
23     function(n);
24     printf("%d",c+2);
25     return 0;
26 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! 

Correct

Marks for this submission: 1.00/1.00.

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Started on	Wednesday, 13 August 2025, 10:25 AM
State	Finished
Completed on	Wednesday, 13 August 2025, 11:28 AM
Time taken	1 hour 2 mins
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num; ++i)
    {
        if (num % i == 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include <stdio.h>
2  int c=0;
3  void Factor(int num) {
4  {
5      for (int i = 1; i <= num; ++i)
6      {
7          c++;
8          c++;
9          if (num % i == 0)
10         {
11             c++;
12         }
13     }
14 }
15 }
16 }
17 int main()
18 {
19     int num;
20     scanf("%d",&num);
21     Factor(num);
22     printf("%d",c+1);
23     return 0;
24 }
25
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓

	Input	Expected	Got	
✓	4	12	12	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Wednesday, 13 August 2025, 10:24 AM
State	Finished
Completed on	Wednesday, 13 August 2025, 11:18 AM
Time taken	53 mins 50 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("");
                printf("");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include <stdio.h>
2  int c=0;
3  void func(int n)
4  {
5      if(n==1)
6      {
7          c++;
8      }
9      else
10     {
11         c++;
12         for(int i=1; i<=n; i++)
13         {
14             c++;
15             for(int j=1; j<=n; j++)
16             {
17                 c++;
18                 c++;
19                 c++;
20                 c++;
21                 break;
22             }
23         }
24     }
25 }
26 int main()
27 {
28     int n;
```

```
29 | scanf("%d",&n);
30 | func(n);
31 | printf("%d",c+1);
32 | return 0;
33 | }
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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S2

Started on	Wednesday, 13 August 2025, 10:46 AM
State	Finished
Completed on	Wednesday, 13 August 2025, 11:09 AM
Time taken	22 mins 26 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  int c=0;
3  void function (int n)
4  {
5      int i= 1;
6      c++;
7      int s =1;
8      c++;
9      while(s <= n)
10     {
11         c++;
12         i++;
13         c++;
14         s += i;
15         c++;
16     }
17     printf("%d",c+1);
18 }
19 int main()
20 {
21     int n;
22     scanf("%d", &n);
23     function(n);
24     return 0;
25 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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